KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5283C | 5 Credits: 4 | Program Elective | Source-grounded

Internet and Web Application

After completion of this course the student will be able to:

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5283C is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5283C.pdf from the uploaded official SITTTR syllabus archive.
Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Cyber Forensics and Information Security
Course code	5283C
Course title	Internet and Web Application
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

18 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- After completion of this course the student will be able to:
- Reveal the underlying in web application.
- Understand the security principles in developing a reliable web application.
- Understand security concepts, security professional roles, and security resources
- in the context of systems and security development life cycle.

Course outcomes

CO	Official outcome	Study level
CO1	Understand Internet security and applications. 16 Understanding Understand web application security fundamentals and	See official syllabus
CO2	14 Understanding browser security principles.	See official syllabus
CO3	Understand Web Application Vulnerabilities. 12 Understanding Exposure to various security threats to web applications/	See official syllabus
CO4	16 Understanding servers and providing security to web servers. SeriesTest 2	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2

Row	Extracted official mapping
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Understand Internet security and applications.	5	As specified
Module 2	principles.	6	As specified
Module 3	To understand Web Application Vulnerabilities.	4	As specified
Module 4	security to web servers.	3	As specified

MODULE 1

Understand Internet security and applications.

This module is presented from the official syllabus scope for Internet and Web Application. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand Internet security and applications.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ 4 Understanding Authentication & Authorization

4 Understanding Authentication & Authorization is an official topic in Module 1 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding Authentication & Authorization using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding authentication & authorization should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding Authentication & Authorization means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding Authentication & Authorization □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Firewalls, Encryption & Decryption, SSL 4 Understanding Internet Application - E-Mail, Email protocols,

Firewalls, Encryption & Decryption, SSL 4 Understanding Internet Application - E-Mail, Email protocols, is an official topic in Module 1 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Firewalls, Encryption & Decryption, SSL 4 Understanding Internet Application - E-Mail, Email protocols, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, firewalls, encryption & decryption, ssl 4 understanding internet application - e-mail, email protocols, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Firewalls, Encryption & Decryption, SSL 4 Understanding Internet Application - E-Mail, Email protocols, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Firewalls, Encryption & Decryption, SSL 4 Understanding Internet Application - E-Mail, Email protocols, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ 2 Understanding Telnet

2 Understanding Telnet is an official topic in Module 1 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Telnet using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding telnet should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Telnet means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 2 Understanding Telnet □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▾ FTP, Newsgroup, Chart room

FTP, Newsgroup, Chart room is an official topic in Module 1 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify FTP, Newsgroup, Chart room using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ftp, newsgroup, chart room should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What FTP, Newsgroup, Chart room means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] FTP, Newsgroup, Chart room [illegible] definition/meaning [illegible]. [illegible], steps, application [illegible]. Technical terms English-[illegible].

▼ Internet Relay Chat, Video Conferencing 3 Understanding Contents :Introduction to internet security, Types of security, Authentication and authorization, Firewalls, Encryption & Decryption, SSL, Internet Application - E-Mail, Email protocols, Telnet, FTP, Newsgroup, Chart room, Internet Relay Chat, Video Conferencing Understand web application security fundamentals and browser security

Internet Relay Chat, Video Conferencing 3 Understanding Contents :Introduction to internet security, Types of security, Authentication and authorization, Firewalls, Encryption & Decryption, SSL, Internet Application - E-Mail, Email protocols, Telnet, FTP, Newsgroup, Chart room, Internet Relay Chat, Video Conferencing Understand web application security fundamentals and browser security is an official topic in Module 1 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Internet Relay Chat, Video Conferencing 3 Understanding Contents :Introduction to internet security, Types of security, Authentication and authorization, Firewalls, Encryption & Decryption, SSL, Internet Application - E-Mail, Email protocols, Telnet, FTP, Newsgroup, Chart room, Internet Relay Chat, Video Conferencing Understand web application security fundamentals and browser security using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, internet relay chat, video conferencing 3 understanding contents :introduction to internet security, types of security, authentication and authorization, firewalls, encryption & decryption, ssl, internet application - e-mail, email protocols, telnet, ftp, newsgroup, chart room, internet relay chat, video conferencing understand web application security fundamentals and browser security should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Internet Relay Chat, Video Conferencing 3 Understanding Contents :Introduction to internet security, Types of security, Authentication and authorization, Firewalls, Encryption & Decryption, SSL, Internet Application - E-Mail, Email protocols, Telnet, FTP, Newsgroup, Chart room, Internet Relay Chat, Video Conferencing Understand web application security fundamentals and browser security means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Internet Relay Chat, Video Conferencing 3 Understanding Contents :Introduction to internet security, Types of security, Authentication and authorization, Firewalls, Encryption & Decryption, SSL, Internet Application - E-Mail, Email protocols, Telnet, FTP, Newsgroup, Chart room, Internet Relay Chat, Video Conferencing Understand web application security fundamentals and browser security □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

principles.

This module is presented from the official syllabus scope for Internet and Web Application. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: principles.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ Input Validation

Input Validation is an official topic in Module 2 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Input Validation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, input validation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Input Validation means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Input Validation □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▾ Attack Surface Reduction Rules of Thumb

Attack Surface Reduction Rules of Thumb is an official topic in Module 2 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Attack Surface Reduction Rules of Thumb using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, attack surface reduction rules of thumb should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Attack Surface Reduction Rules of Thumb means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Attack Surface Reduction Rules of Thumb □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Classifying and Prioritizing Threads

Classifying and Prioritizing Threads is an official topic in Module 2 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classifying and Prioritizing Threads using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classifying and prioritizing threads should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Classifying and Prioritizing Threads means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Classifying and Prioritizing Threads □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Origin Policy - Exceptions to the Same-Origin Policy

Origin Policy - Exceptions to the Same-Origin Policy is an official topic in Module 2 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Origin Policy - Exceptions to the Same-Origin Policy using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, origin policy - exceptions to the same-origin policy should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Origin Policy - Exceptions to the Same-Origin Policy means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Origin Policy - Exceptions to the Same-Origin Policy [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ Cross-Site Scripting and Cross-Site Request Forgery

Cross-Site Scripting and Cross-Site Request Forgery is an official topic in Module 2 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Cross-Site Scripting and Cross-Site Request Forgery using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, cross-site scripting and cross-site request forgery should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Cross-Site Scripting and Cross-Site Request Forgery means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Cross-Site Scripting and Cross-Site Request Forgery □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Reflected XSS - HTML Injection 3 Understanding SeriesTest–I 1 Prioritizing Threads, Origin Policy - Exceptions to the Same-Origin Policy - Cross-Site Scripting and Cross-Site Request Forgery - Reflected XSS - HTML Injection

Reflected XSS - HTML Injection 3 Understanding SeriesTest–I 1 Prioritizing Threads, Origin Policy - Exceptions to the Same-Origin Policy - Cross-Site Scripting and Cross-Site Request Forgery - Reflected XSS - HTML Injection is an official topic in Module 2 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Reflected XSS - HTML Injection 3 Understanding SeriesTest–I 1 Prioritizing Threads, Origin Policy - Exceptions to the Same-Origin Policy - Cross-Site Scripting and Cross-Site Request Forgery - Reflected XSS - HTML Injection using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, reflected xss - html injection 3 understanding seriestest–i 1 prioritizing threads, origin policy - exceptions to the same-origin policy - cross-site scripting and cross-site request forgery - reflected xss - html injection should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Reflected XSS - HTML Injection 3 Understanding SeriesTest–I 1 Prioritizing Threads, Origin Policy - Exceptions to the Same-Origin Policy - Cross-Site Scripting and Cross-Site Request Forgery - Reflected XSS - HTML Injection means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Reflected XSS - HTML Injection 3 Understanding SeriesTest–I 1 Prioritizing Threads, Origin Policy - Exceptions to the Same-Origin Policy - Cross-Site Scripting and Cross-Site Request Forgery - Reflected XSS - HTML Injection [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

To understand Web Application Vulnerabilities.

This module is presented from the official syllabus scope for Internet and Web Application. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: To understand Web Application Vulnerabilities.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ SQL injection, cross domain attack 4 Understanding (XSS/XSRF/XSSI), http header injection

SQL injection, cross domain attack 4 Understanding (XSS/XSRF/XSSI), http header injection is an official topic in Module 3 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify SQL injection, cross domain attack 4 Understanding (XSS/XSRF/XSSI), http header injection using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, sql injection, cross domain attack 4 understanding (xss/xsrf/xssi), http header injection should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What SQL injection, cross domain attack 4 Understanding (XSS/XSRF/XSSI), http header injection means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ SQL injection, cross domain attack 4 Understanding (XSS/XSRF/XSSI), http header injection □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ SSL vulnerabilities and testing

SSL vulnerabilities and testing is an official topic in Module 3 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify SSL vulnerabilities and testing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ssl vulnerabilities and testing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What SSL vulnerabilities and testing means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ SSL vulnerabilities and testing □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ Proper encryption use in web application. 2 Understanding Session vulnerabilities and testing – Common

Proper encryption use in web application. 2 Understanding Session vulnerabilities and testing – Common is an official topic in Module 3 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Proper encryption use in web application. 2 Understanding Session vulnerabilities and testing – Common using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, proper encryption use in web application. 2 understanding session vulnerabilities and testing – common should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	**What to prepare**
Definition/identity	What Proper encryption use in web application. 2 Understanding Session vulnerabilities and testing – Common means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Proper encryption use in web application. 2 Understanding Session vulnerabilities and testing – Common □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▾ 3 Applying website weakness Contents : Client state manipulation, cookie based attacks, SQL injection, , cross domain attack (XSS/XSRF/XSSI) http header injection, SSL vulnerabilities and testing – Proper encryption use in web application – Session vulnerabilities and testing - Common website weakness. Exposure to various security threats to web applications/ servers and providing

3 Applying website weakness Contents : Client state manipulation, cookie based attacks, SQL injection, , cross domain attack (XSS/XSRF/XSSI) http header injection, SSL vulnerabilities and testing –Proper encryption use in web application – Session vulnerabilities and testing - Common website weakness. Exposure to various security threats to web applications/ servers and providing is an official topic in Module 3 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying website weakness Contents : Client state manipulation, cookie based attacks, SQL injection, , cross domain attack (XSS/XSRF/XSSI) http header injection, SSL vulnerabilities and testing – Proper encryption use in web application – Session vulnerabilities and testing - Common website weakness. Exposure to various security threats to web applications/ servers and providing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying website weakness contents : client state manipulation, cookie based attacks, sql injection, , cross domain attack (xss/xsrf/xssi) http header injection, ssl vulnerabilities and testing –proper encryption use in web application – session vulnerabilities and testing - common website weakness. exposure to various security threats to web applications/ servers and providing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying website weakness Contents : Client state manipulation, cookie based attacks, SQL injection, , cross domain attack (XSS/XSRF/XSSI) http header injection, SSL vulnerabilities and testing – Proper encryption use in web application – Session vulnerabilities and testing - Common website weakness. Exposure to various security threats to web applications/ servers and providing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 3 Applying website weakness Contents : Client state manipulation, cookie based attacks, SQL injection, , cross domain attack (XSS/XSRF/XSSI) http header injection, SSL vulnerabilities and testing –Proper encryption use in web application – Session vulnerabilities and testing - Common website weakness. Exposure to various security threats to web applications/ servers and providing [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

security to web servers.

This module is presented from the official syllabus scope for Internet and Web Application. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: security to web servers.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 6 Understanding mechanisms

6 Understanding mechanisms is an official topic in Module 4 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding mechanisms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding mechanisms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding mechanisms means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 6 Understanding mechanisms □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Illustrate the mechanism of transmitting data 4 Understanding Explain vulnerabilities of access control and

Illustrate the mechanism of transmitting data 4 Understanding Explain vulnerabilities of access control and is an official topic in Module 4 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the mechanism of transmitting data 4 Understanding Explain vulnerabilities of access control and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the mechanism of transmitting data 4 understanding explain vulnerabilities of access control and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the mechanism of transmitting data 4 Understanding Explain vulnerabilities of access control and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Illustrate the mechanism of transmitting data 4 Understanding Explain vulnerabilities of access control and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 6 Understanding securing mechanisms. SeriesTest–II 1 Contents :Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms –securing authentication Attacking access controls – Common vulnerabilities – Securing access controls. Text/Reference T/R Book Title/Author Sullivan, Bryan and Vincent Liu Web Application Security A Beginner's Guide T2 McGraw Hill Professional 2011. Stuttard, Dafydd and Marcus Pinto The Web Application Hacker's Handbook: T3 Finding and Exploiting Security Flaws John Wiley Sons 2011 R1 Mike Harwood, Ron Price Internet and Web Application Security, 3rd Edition

6 Understanding securing mechanisms. SeriesTest–II 1 Contents :Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms –securing authentication Attacking access controls – Common vulnerabilities – Securing access controls. Text/Reference T/R Book Title/Author Sullivan, Bryan and Vincent Liu Web Application Security A Beginner's Guide T2 McGraw Hill Professional 2011. Stuttard, Dafydd and Marcus Pinto The Web Application Hacker's Handbook: T3 Finding and Exploiting Security Flaws John Wiley Sons 2011 R1 Mike Harwood, Ron Price Internet and Web Application Security, 3rd Edition is an official topic in Module 4 of Internet and Web Application. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding securing mechanisms. SeriesTest–II 1 Contents :Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms –securing authentication Attacking access controls – Common vulnerabilities – Securing access controls. Text/Reference T/R Book Title/Author Sullivan, Bryan and Vincent Liu Web Application Security A Beginner’s Guide T2 McGraw Hill Professional 2011. Stuttard, Dafydd and Marcus Pinto The Web Application Hacker’s Handbook: T3 Finding and Exploiting Security Flaws John Wiley Sons 2011 R1 Mike Harwood, Ron Price Internet and Web Application Security, 3rd Edition using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding securing mechanisms. seriestest–ii 1 contents :web application security- key problem factors – core defense mechanisms- handling user access- handling user input- handling attackers – web spidering – discovering hidden content. transmitting data via the client – hidden form fields – http cookies – url parameters – handling client-side data securely – attacking authentication – design flaws in authentication mechanisms –securing authentication attacking access controls – common vulnerabilities –

securing access controls. text/reference t/r book title/author sullivan, bryan and vincent liu web application security a beginner’s guide t2 mcgraw hill professional 2011. stuttard, dafydd and marcus pinto the web application hacker’s handbook: t3 finding and exploiting security flaws john wiley sons 2011 r1 mike harwood, ron price internet and web application security, 3rd edition should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding securing mechanisms. SeriesTest–II 1 Contents :Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms –securing authentication Attacking access controls – Common vulnerabilities – Securing access controls. Text/Reference T/R Book Title/Author Sullivan, Bryan and Vincent Liu Web Application Security A Beginner’s Guide T2 McGraw Hill Professional 2011. Stuttard, Dafydd and Marcus Pinto The Web Application Hacker’s Handbook: T3 Finding and Exploiting Security Flaws John Wiley Sons 2011 R1 Mike Harwood, Ron Price Internet and Web Application Security, 3rd Edition means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 6 Understanding securing mechanisms. SeriesTest–II 1 Contents :Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms –securing authentication Attacking access controls – Common vulnerabilities – Securing access controls. Text/Reference T/R Book Title/Author Sullivan, Bryan and Vincent Liu Web Application Security A Beginner’s Guide T2 McGraw Hill Professional 2011. Stuttard, Dafydd and Marcus Pinto The Web Application Hacker’s Handbook: T3 Finding and Exploiting Security Flaws John Wiley Sons 2011 R1 Mike Harwood, Ron Price Internet and Web Application Security, 3rd Edition □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain 4 Understanding Authentication & Authorization. (3 marks)**

▶ **Define or explain Firewalls, Encryption & Decryption, SSL 4 Understanding Internet Application - E-Mail, Email protocols,. (3 marks)**

▶ **Define or explain 2 Understanding Telnet. (3 marks)**

▶ **Define or explain FTP, Newsgroup, Chart room. (3 marks)**

Module 2

▶ **Define or explain Input Validation. (3 marks)**

▶ **Define or explain Attack Surface Reduction Rules of Thumb. (3 marks)**

▶ **Define or explain Classifying and Prioritizing Threads. (3 marks)**

▶ **Define or explain Origin Policy - Exceptions to the Same-Origin Policy. (3 marks)**

Module 3

▶ **Define or explain SQL injection, cross domain attack 4 Understanding (XSS/XSRF/XSSI), http header injection. (3 marks)**

▶ **Define or explain SSL vulnerabilities and testing. (3 marks)**

▶ **Define or explain Proper encryption use in web application. 2 Understanding Session vulnerabilities and testing – Common. (3 marks)**

▶ **Define or explain 3 Applying website weakness Contents : Client state manipulation, cookie based attacks, SQL injection, , cross domain attack (XSS/XSRF/XSSI) http header injection, SSL vulnerabilities and testing – Proper encryption use in web application – Session vulnerabilities and testing - Common website weakness. Exposure to various security threats to web applications/ servers and providing. (3 marks)**

Module 4

▶ **Define or explain 6 Understanding mechanisms. (3 marks)**

▶ **Define or explain Illustrate the mechanism of transmitting data 4 Understanding Explain vulnerabilities of access control and. (3 marks)**

▶ **Define or explain 6 Understanding securing mechanisms. SeriesTest–II 1 Contents :Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms –securing authentication Attacking access controls – Common vulnerabilities – Securing access controls. Text/Reference T/R Book Title/Author Sullivan, Bryan and Vincent Liu Web Application Security A Beginner’s Guide T2 McGraw Hill Professional 2011. Stuttard, Dafydd and Marcus Pinto The Web Application Hacker’s Handbook: T3 Finding and Exploiting Security**

Flaws John Wiley Sons 2011 R1 Mike Harwood, Ron Price Internet and Web Application Security, 3rd Edition. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **4 Understanding Authentication & Authorization**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Input Validation**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **SQL injection, cross domain attack 4 Understanding (XSS/XSRF/XSSI), http header injection**.

3-mark explanation: Explain one principle, process or comparison from

Module 4

1-mark definition: State the meaning of **6 Understanding mechanisms**.

3-mark explanation: Explain one principle, process or comparison from this module.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

No separate web-resource heading was detected in the extracted text.

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 5283C. Verify institutional updates against the current official curriculum.